Customer Churn Analysis and Customer Intelligence

End-to-end Data Analytics Portfolio Report | SQLite + Python + Pandas + Matplotlib + Seaborn

21	28.6%	71.4%	Rs 73.94
Customers	Churn Rate	Retention	MRR at Risk

Executive Summary

This project analyzes customer churn for a small OTT-style subscription dataset by combining customer, subscription, and support information. The workflow extracts data from SQLite, cleans and standardizes fields in pandas, engineers a binary churn flag and risk tiers, performs grouped exploratory analysis, and converts the results into business-focused recommendations. The project contains **21 customers across 3 relational tables** and identifies **6 churned customers**, producing an overall churn rate of **28.6%**.

The clearest retention signal is contract structure: **monthly subscribers churn at 55.6%** versus **8.3%** for annual subscribers. Churned customers represent **Rs 73.94** of monthly charges, while their combined CLTV is **Rs 2,047**. The notebook also records a strong positive correlation of **0.77** between escalation and churn; this is an association in a very small sample and should not be interpreted as causal.

> This report follows the business-oriented structure of the supplied reference report - business problem, data pipeline, KPI analysis, visual insights and action items - while using the actual outputs from the submitted notebook and exported dataset.

1. Business Problem

Customer churn reduces recurring revenue, increases acquisition pressure and can expose weaknesses in pricing, service quality, product experience or customer support. The objective of this analysis is to answer three practical questions:

- **Who?** Which customers or customer segments are most exposed to churn?
- **Why?** What contract, plan, geography, acquisition/subscription type and support patterns appear alongside churn?
- **When?** Do cancellations cluster in particular months or periods?

Business objective

Quantify churn, isolate high-risk segments, estimate monthly revenue exposure and surface operational actions that a retention team could prioritize.

2. Project Objectives

- Integrate multiple customer-related tables from SQLite into a single analytical dataset.
- Clean data types, standardize categorical values, remove unusable fields and repair missing country values.
- Create churn and risk-related features for downstream analysis.
- Calculate core KPIs and segment churn by plan, contract, state and subscription type.
- Analyze support escalation and churn relationship.
- Translate analytical results into practical retention recommendations.

3. Technology Stack

Technology	Role
SQLite / sqlite3	Relational storage, table discovery, schema inspection and data extraction
Python	Primary analytical environment
Pandas	Data cleaning, merging, grouping, pivot analysis and export
NumPy	Conditional feature engineering and numeric transformations
Matplotlib	Business charts and time-series visualization
Seaborn	Correlation heatmap, pairplot and categorical analysis
Jupyter Notebook	Reproducible project workflow and documentation

4. Data Source and Relational Model

The source database is **customer_churn** and contains three tables linked by **customerid**. The notebook first queries SQLite for table names and then inspects schemas before loading each table into pandas.

Table	Key fields	Purpose
db_customer	customerid, country, state, gender, dob	Customer profile and demographic information
db_subscription	customerid, plan_type, contract_type, cancellation_date, monthly_charges, cltv, churn_score	Subscription lifecycle, commercial value and churn-related fields
db_support	customerid, complaint_date, escalations, csat_score	Support interaction and escalation signals

Dataset scale

Customer table: 21 records. Subscription table: 21 records. Support table: 9 support records across 7 distinct customers before deduplication. After counting complaints and keeping the latest support row per customer, the analytical dataset contains 21 customer rows and 21 columns.

5. Data Preparation

The notebook applies the following preparation steps:

- Renamed **name** to **customer_name** for readability.
- Dropped **interests** and **pincode** from the customer table because they were not used in the analysis; pincode was completely null in the raw file.
- Converted DOB, subscription dates, renewal dates, cancellation dates and complaint dates to datetime.
- Standardized gender labels from **Men/Women** to **Male/Female**.
- Filled missing country values using the state-to-country mapping derived from non-null records.
- Created **complaint_count** per customer, then sorted support records by complaint date and retained the latest support record per customer.
- Merged subscription + customer + support data using left joins on customerid.

Quality check

The notebook explicitly checks dataset shape before and after the support cleanup and merge. The first merge created 23 rows because support contained repeated customerids; after deduplicating support to one row per customer, the final dataset returned to the expected 21 rows.

6. Feature Engineering

Churn flag: 1 when cancellation_date is present; otherwise 0. This produces 6 churned customers out of 21. **Customer tenure:** calculated from subscription_start_date to cancellation_date for churned customers, or to the current execution date for active customers. **Churn risk:** score-based tiers - Low (<50), Medium (50-69), High (>=70).

Important analytical note

The churn_score is an existing field in the dataset. The notebook creates risk tiers from it but does not train a machine-learning model. Because the score may already encode churn-related information, it should not be described as a newly built predictive model in a resume or portfolio description.

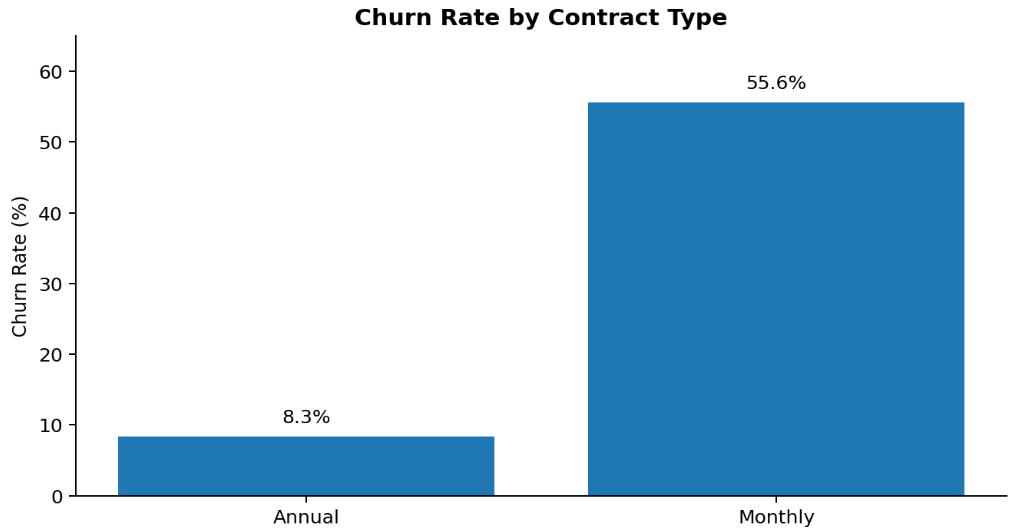
7. KPI Results

KPI	Result	Interpretation
Total customers	21	Small analytical sample
Churned customers	6	Cancellation date present
Churn rate	28.57%	Share of customers who churned
Retention rate	71.43%	Share not flagged as churned
ARPU	Rs 18.85	Average monthly charge per customer
Total monthly charges	Rs 395.79	Sum of monthly charges in dataset
Revenue at risk	Rs 73.94	Monthly charges attached to churned customers
Revenue loss share	18.7%	Revenue at risk as a share of total monthly charges
CLTV lost	Rs 2,047	Sum of CLTV for churned customers
Avg tenure	1,534 days	Dynamic metric based on notebook execution date
Escalation rate	19.05%	Rows with escalation = Y
Avg complaints / customer	0.43	Calculated from complaint_count
Escalation-churn correlation	0.77	Strong positive association in this sample

8. Segment Analysis

Contract type

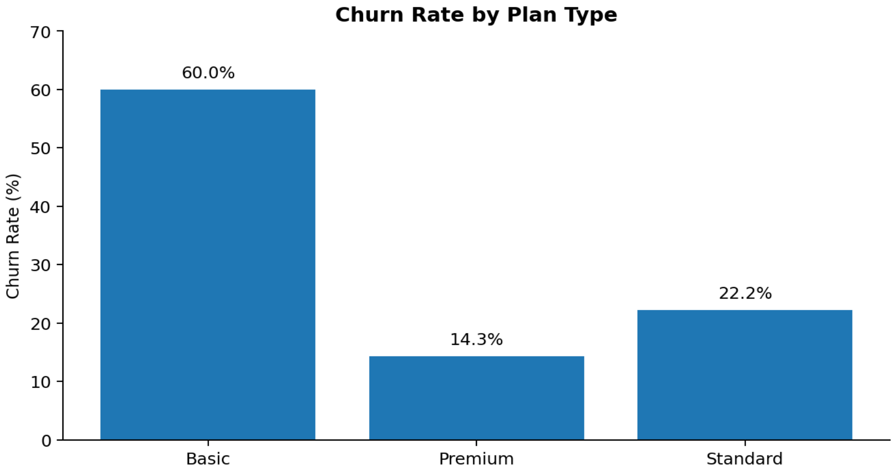
Contract	Customers	Churned	Churn Rate
Annual	12	1	8.3%
Monthly	9	5	55.6%



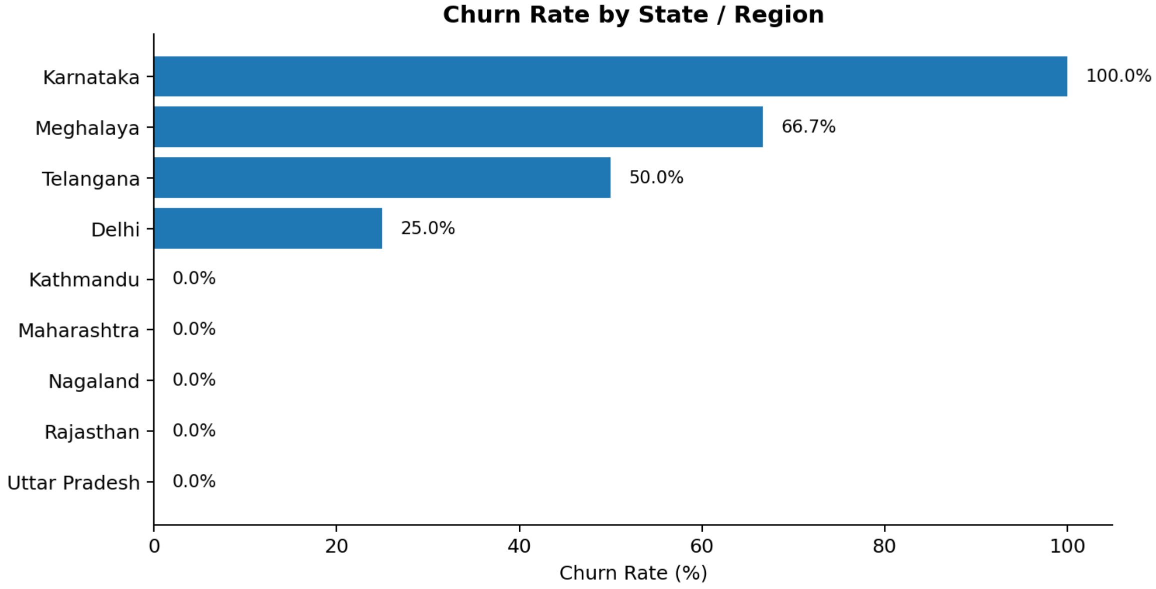
Key observation: monthly contracts are the dominant retention risk. Churn is 55.6% for monthly contracts compared with 8.3% for annual contracts - a 47.2 percentage-point gap.

Plan type

Plan	Customers	Churned	Churn Rate
Basic	5	3	60.0%
Premium	7	1	14.3%
Standard	9	2	22.2%


Churn Rate by Plan Type
Churn Rate (%)
70
60
50
40
30
20
10
0
60.0%
14.3%
22.2%
Basic
Premium
Standard

9. Geographic and Acquisition / Subscription Analysis



Geographic highlights

- Karnataka shows 100% churn (2 of 2 customers) in this sample.
- Meghalaya shows 66.7% churn (2 of 3).
- Telangana shows 50.0% churn (1 of 2).
- Delhi shows 25.0% churn (1 of 4).
- Other observed regions have 0 churn in the current sample.

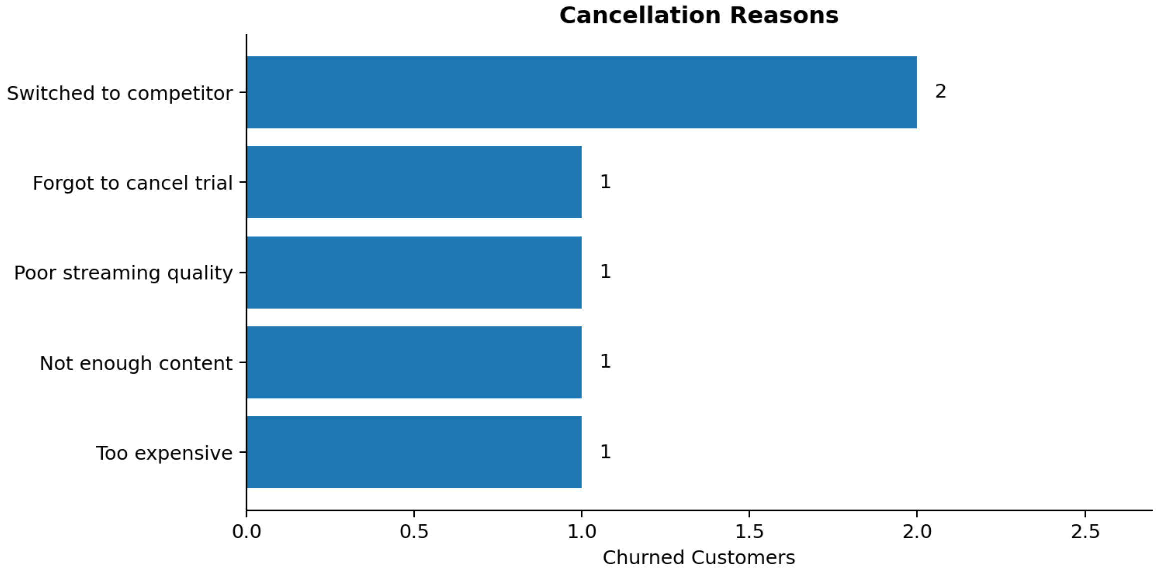
These percentages are useful for investigation, but the counts are very small. A larger customer population is required before making geographic policy decisions.

Subscription / acquisition type

Type	Customers	Churned	Churn Rate	Monthly charges
Organic	9	0	0.0%	Rs 145.91
Paid	6	1	16.7%	Rs 174.94
Referral	6	5	83.3%	Rs 74.94

Referral-labeled customers (dataset spelling: *Refferal*) have the highest churn rate at 83.3% (5 of 6), while Organic has 0% churn in this sample. This is a strong signal for deeper acquisition-quality analysis, but not enough evidence by itself to change channel strategy.

10. Cancellation and Support Intelligence



Cancellation reasons

- Switched to competitor: 2 customers (33.3% of churned customers).
- Too expensive: 1 customer.
- Not enough content: 1 customer.
- Poor streaming quality: 1 customer.
- Forgot to cancel trial: 1 customer.

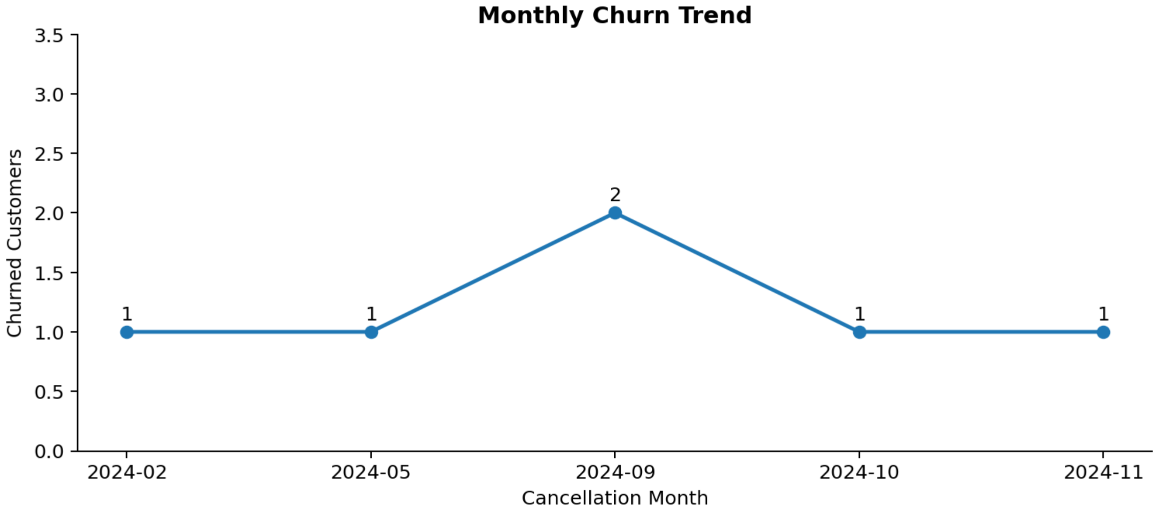
Support signal

The notebook reports an escalation-churn correlation of **0.77**. In the cleaned customer-level dataset, all 4 customers with an escalation flag of Y are churned, while 2 of 17 customers without an escalation flag are churned. This is a compelling operational signal, but the dataset is too small for causal inference or statistical generalization.

Support interpretation

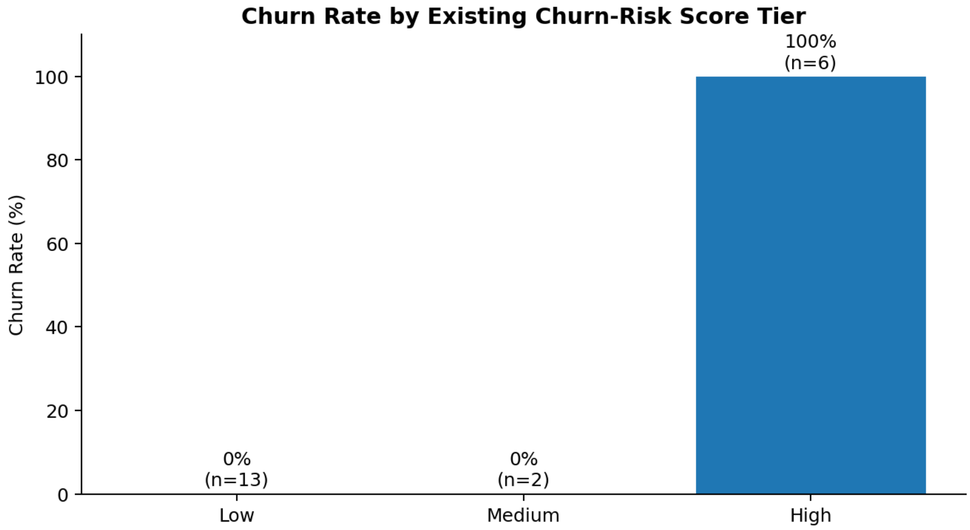
- Escalated cases should be treated as a retention-alert signal.
- Low CSAT appears alongside several churned support interactions, suggesting service recovery may matter.
- A production version should aggregate support history at customer level instead of keeping only the latest support row.

11. Churn Timing and Risk Segmentation



Timing insight

Six churn events occur between February and November 2024, with the highest monthly count occurring in September 2024 (2 churns). The project therefore provides a starting point for investigating whether pricing, service quality, renewal events, content changes or competitor activity coincided with that period.



Risk-tier result

The existing churn_score classifies 13 customers as Low, 2 as Medium and 6 as High. In this dataset, all 6 High-risk customers are churned, while Low and Medium groups show 0 churn. Because the score is a pre-existing field, this result demonstrates how the field can be operationalized - not that a new predictive model was built.

12. Business Recommendations

- **Encourage monthly-to-annual migration:** test annual-plan incentives, renewal benefits or bundled value propositions for monthly customers.
- **Create an escalation retention workflow:** route escalated support cases to proactive retention teams, with SLA-based follow-up and service recovery.
- **Investigate Karnataka and September 2024:** review pricing changes, technical incidents, complaints, content changes and competitor activity.

- **Address cancellation drivers:** combine competitor analysis, pricing/value review and product-quality improvements.
- **Prioritize by customer value:** combine churn risk with CLTV so retention effort is concentrated where expected revenue impact is highest.

13. Project Workflow

Stage	Implementation
1. Connect	Python sqlite3 connection to customer_churn database
2. Extract	SQLite queries for table discovery and pandas read_sql for imports
3. Clean	Rename/drop fields, datetime conversion, standardization, missing-value repair
4. Engineer	churn_flag, complaint_count, tenure_days and churn_risk
5. Join	Left-merge customer, subscription and support tables on customerid
6. Analyze	KPI calculation, groupby, aggregation and pivot tables
7. Visualize	Matplotlib + Seaborn charts
8. Export	Final analytical dataset exported to CSV
9. Communicate	Business findings translated into actionable retention recommendations

14. Limitations and Next Steps

- The dataset contains only 21 customers, so percentages can be unstable and should be treated as directional.
- The project is descriptive/diagnostic analytics; it does not train or validate a predictive churn model.
- The churn_score is an existing feature, so its use introduces potential target leakage if used as a model input without careful definition.
- The support table is reduced to the latest row per customer for the final merge. A production design should aggregate the full complaint history to avoid losing historical support signals.
- Tenure for active customers depends on the notebook execution date, so the metric changes over time.
- Next improvement: build a larger time-aware dataset, create customer-level support aggregates, validate statistical significance and add a calibrated ML model only after preventing leakage.

15. GitHub Portfolio Positioning

This project demonstrates a complete analyst workflow: relational data extraction, Python-SQL integration, data cleaning, feature engineering, exploratory analysis, KPI design, visualization and business recommendations. It is stronger as a portfolio piece when the repository clearly separates raw data, analytical code, visuals and documentation.

Suggested repository structure

```
churn-analysis/
|-- churn_analysis.ipynb
|-- customer_churn.db
|-- exported_churn_data.csv
|-- customer_churn_data_raw.xlsx
|-- README.md
|-- CHURN_ANALYSIS_REPORT.pdf
`-- assets/
    |-- churn_by_contract.png
    |-- churn_by_plan.png
    |-- churn_by_state.png
    |-- cancellation_reasons.png
    |-- monthly_churn_trend.png
    `-- risk_tiers.png
```

Conclusion

The analysis finds an overall churn rate of **28.6%**, with the largest structural gap between monthly and annual contracts. The churned cohort represents **Rs 73.94** of monthly charges and **Rs 2,047** in combined CLTV. The findings support a practical retention strategy centered on contract migration, escalated-support intervention, investigation of high-churn segments and value-based prioritization.

This report is based on the submitted project notebook, database, raw spreadsheet and exported analytical dataset.